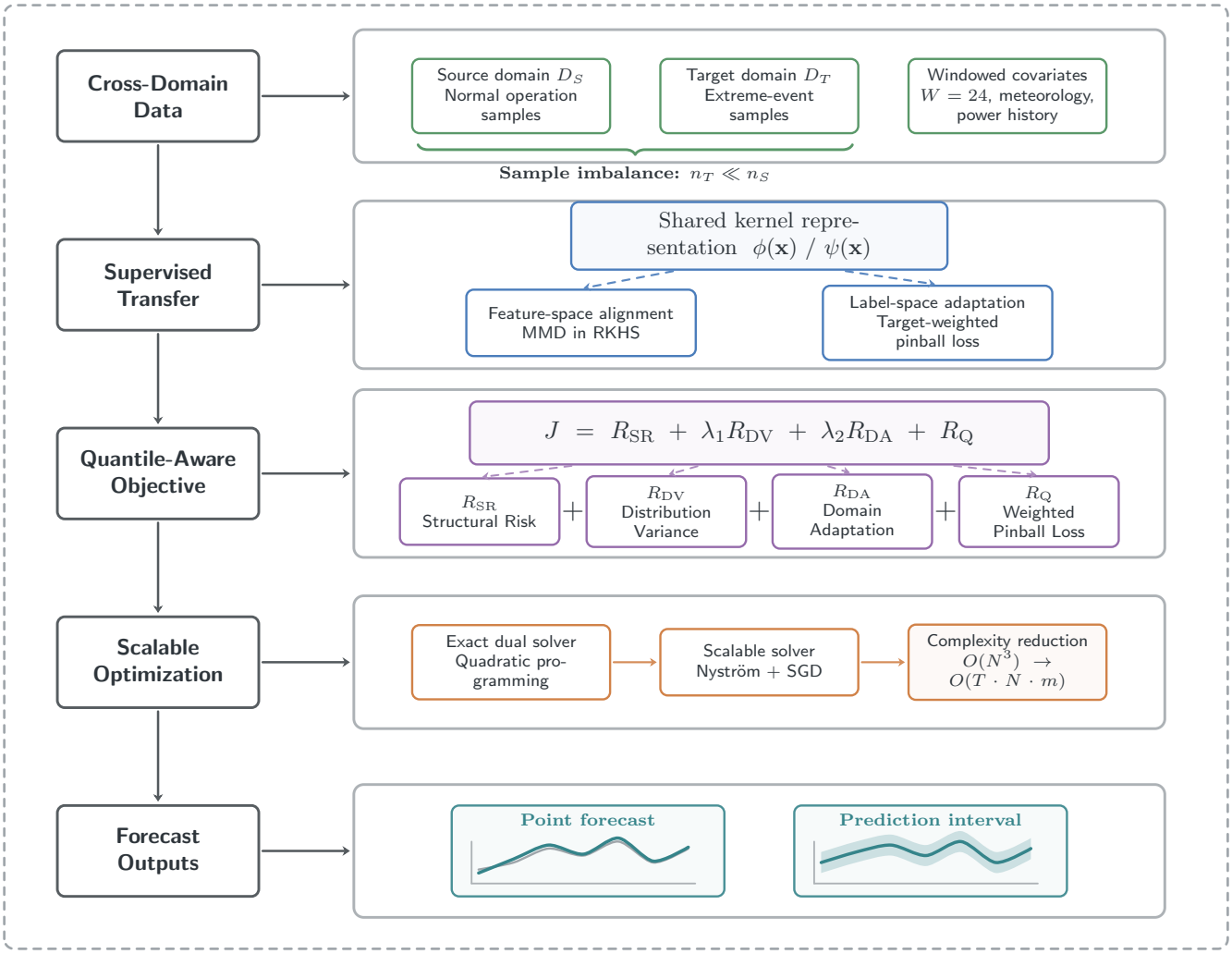

Cross-Domain Data
Source domain D_S Normal operation samples
Target domain D_T Extreme-event samples
Windowed covariates $W = 24$, meteorology, power history
Sample imbalance: $n_T \ll n_S$
Supervised Transfer
Shared kernel representation $\phi(\mathbf{x})$ / $\psi(\mathbf{x})$
Feature-space alignment MMD in RKHS
Label-space adaptation Target-weighted pinball loss
Quantile-Aware Objective
$J = R_{\mathrm{SR}} + \lambda_1 R_{\mathrm{DV}} + \lambda_2 R_{\mathrm{DA}} + R_{\mathrm{Q}}$
R_{SR} Structural Risk
R_{DV} Distribution Variance
R_{DA} Domain Adaptation
R_{Q} Weighted Pinball Loss
Scalable Optimization
Exact dual solver Quadratic programming
Scalable solver Nyström + SGD
Complexity reduction $O(N^3) \rightarrow O(T \cdot N \cdot m)$
Forecast Outputs
Point forecast
Prediction interval